Investigating the Impact of Losses on FGVC Performance and Geometry

Loss-function ablation and species-count scaling ladder on SWIN-Base

Internal draft: results so far

Farid Karimli

October 13, 2026

Abstract

We hold one model and one training recipe fixed (SWIN-Base, 384 px, ImageNet-22k initialisation, 50 epochs on Kaggle Herbarium 2022, 15,501 species) and change only the training loss. We compare eight losses against a plain ArcFace reference, all trained at full scale and all scored on the full 167,955-image validation set. Two losses that give each species its own margin, DBM and KappaFace, clearly win. They beat ArcFace by +1.9 and +1.8 points of macro-F1 and by +1.5 and +1.2 points of top-1 accuracy. A second seed moves any of these scores by about 0.1 points or less. Their gains are largest for species with few training images: up to +6 points of per-species F1 for species with 6–10 images, against about +1 point for species with 51–73. KappaFace is better than DBM below about 10 training images, and DBM is better above about 15. Sub-center ArcFace gives a small but real gain. TARA has no detectable effect. ECC and HACE are slightly worse than ArcFace, and X2-Softmax fails (its features collapse). Tidier average representation geometry does not mean better accuracy: the two best losses have the loosest species clusters of any working loss. On a scaling ladder of nested label sets with 15.5k, 32k and 64k species, at fixed compute, DBM beats ArcFace on every rung. Its macro-F1 lead grows with the number of species (from +2.6 to +3.0 to +3.9 points), while its accuracy lead stays flat (+1.0 to +1.3).

1 Motivation

Identifying the species on a digitised herbarium sheet is a fine-grained problem with a very long label list. Kaggle Herbarium 2022 alone has 15,501 species, and the merged Herbarium 2021 and 2022 data has 64,113 (de Lutio et al., 2021). Two things make it hard.

The remaining mistakes are between close relatives. For our SWIN-Base models, 75–78% of wrong answers still name a species in the correct genus, and 87–90% one in the correct family (Section 5.2). By chance those rates would be about 0.3% and 3%. So the model already knows roughly where a specimen belongs. What it gets wrong is separating sister species, often from small differences in leaves, flowers or fruit on a pressed sheet.

Many species have very few examples. In Herbarium 2022 a species has between 2 and 73 training images. In the 64k-species set the median falls to 10 training

Table 1: **Shared training recipe.** Identical for every loss in the ablation. The ladder runs (Section 4) use the same recipe with a fixed step budget in place of the epoch count.

Backbone	SWIN-Base, `microsoft/swin-base-patch4-window12-384-in22k`
Input	384×384, RandomResizedCrop + horizontal flip (no mixup, no advanced aug.)
Train / val data	Kaggle Herbarium 2022: 671,817 / 167,955 images
Label	`scientificNameEncoded`, 15,501 species (15,334 present in val)
Optimiser	AdamW (fused), lr 10^{-4}, weight decay 0.01
Schedule	500 warmup steps, cosine decay over 50 epochs (= 131,250 steps)
Batch	effective 256 (64×2 accum.$\times 2$ GPUs, or $64 \times 4 \times 1$ GPU)
Precision / seed	bf16 / 42 (seed 43 for the noise checks)
Head scale	$s = 30$ for every margin head
Evaluation	full validation set, top-1, top-5 and macro-F1 over present species
Hardware	1–2 NVIDIA L40S on BU SCC; 12 h jobs resumed from checkpoints

images and 3 validation images per species. Macro-F1 weights every species equally, so it is the measure most sensitive to how well these rare species are learned. We treat it as our primary metric and report top-1 accuracy alongside.

An earlier controlled experiment at 224 px showed that replacing cross-entropy (CE) with ArcFace (Deng et al., 2019) changes the feature space much more than the accuracy. The separation gap between species clusters became 16 times wider ($0.012 \rightarrow 0.196$), but top-1 rose by only 1.7 points ($0.7796 \rightarrow 0.7970$). Better class separation in the ArcFace sense therefore did not seem to be the binding constraint, which raised the question this report addresses: *does any loss that intervenes elsewhere, through per-class margins, hierarchy, auxiliary losses or a teacher model, train a better fine-grained classifier? And does the answer hold as the label set grows?*

2 Fixed baseline: what stayed the same

Every run in the ablation uses the same backbone, data, optimiser and schedule. Only the classification head and loss change (Table 1). The reference is ArcFace with one center per class ($s = 30$, $m = 0.30$), applied directly to the 1024-d pooled SWIN feature, with no bottleneck, no BatchNorm and no auxiliary CE head. This is the configuration whose 224 px version gave 0.7971 accuracy and 0.7605 macro-F1 in the earlier controlled CE-vs-ArcFace comparison.

Moving to 384 px forced one change: Swin's attention windows are tied to the input size (Liu et al., 2021), so the 384 px runs use the `swin-base-patch4-window12-384-in22k` checkpoint in place of the 224 px one. At 384 px, ArcFace reaches 0.8299 / 0.7972 (accuracy / macro-F1), 3.3 and 3.7 points above ArcFace at 224 px. Every comparison below is made at 384 px.

Evaluation note. The training script originally scored 2-GPU runs on only the first GPU's half of the validation set, because predictions were never gathered across GPUs. This made macro-F1 about 1 point too low for those runs. We re-scored every

finished model on the full set with one GPU. For the 1-GPU runs this reproduced the logged numbers exactly. All 15k numbers in this report are full-set re-scores, and the ladder runs use the fixed evaluation code.

3 The losses we ablated

The losses change training at four different points: the classifier head and its margin rule, the structure of the output, an extra loss term, or the gradients. We picked them to cover the main ideas for fine-grained, long-tailed recognition: fixed-margin angular losses, per-class adaptive margins, hierarchy-aware outputs, class-center regularisers, and alignment to a biology foundation model. Table 2 summarises them. FlexiGrad, the ninth planned loss, needs multi-task family/genus/species heads and is not yet run.

Throughout, θ_j is the angle between an image's normalised feature and the normalised weight vector of species j, and y is the true species. All margin heads compute logits $s\cos\theta_j$ for the wrong classes and change only the logit of the true class, and only during training. At test time every head predicts with plain $s\cos\theta_j$.

ArcFace (reference) (Deng et al., 2019). The true-class logit becomes $s\cos(\theta_y + m)$ with one fixed margin $m = 0.30$ for every species. To count as correct, an image must be closer to its own species by an extra angle m. We use a 5-epoch margin warmup, which stabilised earlier ArcFace runs. *Why:* ArcFace is the standard angular-margin loss and the strongest single change found so far, so everything else is compared with it.

Sub-center ArcFace ($k = 3$) (Deng et al., 2020). Each species gets three weight vectors, and an image is scored against whichever is closest. *Why:* herbarium species can look very different across sheets (flowering versus fruiting, different preparation), and the extra centers can absorb mislabelled or atypical images.

KappaFace (Oinar et al., 2023). The margin is set per species from two signals: how many training images the species has, and how tightly its features cluster. The tightness is a von Mises–Fisher concentration estimate, recomputed once per epoch from a running mean direction per species. Rare species and spread-out species get a margin near the maximum, $m_0 = 0.30$. Common, tight species get a smaller one. *Why:* it targets exactly the two difficulties of this data, few images and large within-species variation, without any hand-set per-class rule.

DBM, Difficulty-aware Balancing Margin (Son et al., 2025). The margin has two parts. The class part, $m_C = K\,(n_y/n_{\min})^{-\tau}$ with $K = 0.1$ and $\tau = 1$, is fixed by the species' training-image count n_y. It is largest (0.1) for the rarest species and shrinks as $1/n_y$: for a 73-image species in Herbarium 2022 it is about 0.003. The instance part adds $m_C\,(1 - \cos\theta_y)/2$, but only for *hard* images, whose nearest class is not their own. *Why:* it was designed for long-tailed recognition, and it reweights both rare classes and hard examples using only fixed class counts. Every DBM margin is far below ArcFace's 0.30.

Table 2: **Ablated losses and where they act.** n_y is the number of training images of species y. KappaFace and ECC keep per-species running statistics that are not synchronised across GPUs, so they train on one GPU, with the same effective batch.

Loss	Intervenes on	Margin on true class / extra term
ArcFace	head	fixed $m = 0.30$
Sub-center ArcFace	head	fixed $m = 0.30$, 3 centers per species
KappaFace	head	per species, ≤ 0.30, from n_y and feature concentration
DBM	head	per species $0.1\, n_{\min}/n_y$, plus extra for hard images
X2-Softmax	head	quadratic in angle, no margin setting
HACE	output structure	softmax over the full taxonomy tree
ArcFace + ECC	extra loss	pull to own center, push from most similar center
ArcFace + TARA	extra loss	cosine alignment with frozen BioCLIP

X2-Softmax (Xu et al., 2023). The true-class logit is replaced by a quadratic function of the angle, $s\,[a(\theta_y - h)^2 + k]$, with the paper's $a = -1$, $h = -0.3$, $k = 1$. This gives an angle-dependent margin with no margin setting. *Why:* a margin that adapts to how far an image is from its class could suit species of very different difficulty, with one hyperparameter fewer.

HACE, Hierarchy-Aware Cross-Entropy (Chan et al., 2026). The output layer covers the whole taxonomy tree: 15,501 species, 2,564 genera, 272 families and a root, 18,338 nodes in total. A parent node's probability is the sum over its descendants, and the one-hot target is spread up the ancestor path (dilution 0.6). *Why:* most errors are within genus and family, so making the loss aware of the taxonomy could make mistakes milder or rarer.

ArcFace + ECC, Exploration of Class Center (Yao et al., 2024). An extra loss built from a running per-species feature center. It pulls each image toward its own center and pushes it away from the most similar other species' center, more strongly when the two species are more similar. It also adds soft labels from the running average prediction of each species. Loss weights are $\lambda_1 = \lambda_2 = 1$; the paper's values could not be recovered. *Why:* ECC was validated on a Swin-Base backbone for fine-grained classification, the closest match to our setting of any candidate.

ArcFace + TARA (He et al., 2026). An extra loss that aligns a projection of the SWIN feature with the embedding of a frozen BioCLIP (Stevens et al., 2024) image encoder. BioCLIP sees the same random crop at 224 px, and the loss is the negative cosine similarity between the two. This is an adaptation: TARA itself aligns a large multimodal model. *Why:* BioCLIP was trained across the tree of life and may carry taxonomic structure that a classifier trained on herbarium sheets alone does not learn.

4 The scaling ladder

Motivation. The 15k ablation is one label-set size, and the per-species analysis (Section 5.2) suggests that DBM and KappaFace win mainly on rare species. Larger

Table 3: **Ladder label sets.** Nested sets from the merged Herbarium 2021 and 2022 manifest. Larger sets add mostly rare species: the median training images per species falls from 57 to 10.

Rung	Species	Train images	Val images (species in val)	Median images / species	
				train	val
15.5k	15,501	1,366,368	341,932 (15,501)	57	14
32k	32,000	1,667,994	417,176 (29,635)	20	6
64k	64,113	2,249,230	562,308 (57,094)	10	3

herbarium label sets add mostly rare species. So the question is whether the ranking holds, and whether the gap grows, as the number of species goes up.

Construction. We cut three nested label sets from the merged Herbarium 2021 and 2022 manifest (Table 3). The 15.5k set is the 15,501 Herbarium-2022 species. The 32k set adds 16,499 other species drawn at random (seed 0). The 64k set has all 64,113 species. Each larger set contains the smaller ones, and a species keeps the same label id on every rung. The validation set of each rung is restricted to that rung's species.

The 15.5k rung is *not* the same data as the 15k ablation. It has about twice the training images (2021 images of the same species are added), a different validation split, and far more imbalance (up to 1,717 images per species). Ladder numbers are therefore comparable across rungs but not with the 15k ablation tables.

Protocol. We train DBM and ArcFace on every rung, six runs in total, with the recipe of Table 1. Every run gets exactly **131,250 optimizer steps at batch 256**, the same compute as one 15k ablation run. Larger rungs therefore see fewer passes over their data: 24.6, 20.1 and 14.9 epochs. ArcFace on the ladder uses the same registry code path as DBM, so only the head differs. Both are scored every 26,250 steps on the full validation set of their rung, with one seed each. KappaFace ladder runs are in progress and not reported here.

5 Results

5.1 Coarse results on Herbarium 2022 (15.5k species)

Table 4 gives the final full-set scores and Figure 1 the gap of each loss to ArcFace. The intervals come from a paired bootstrap over validation images (1,000 resamples). They capture which images happened to be in the validation set, not seed noise.

- **DBM and KappaFace win, and are nearly tied.** Both beat ArcFace by 1.2–1.5 points of accuracy and 1.8–1.9 points of macro-F1, which is 20–40 times the seed-to-seed spread. DBM is ahead of KappaFace in both seeds, by 0.29 points of accuracy each time and by 0.11 and 0.14 points of macro-F1. The macro-F1 interval of KappaFace minus DBM, $[-0.36, +0.02]$, just includes zero. DBM also has the best top-5 accuracy (0.962), and unlike KappaFace it can train on 2 GPUs.

Table 4: **DBM and KappaFace are the clear winners at 15.5k species.** Final scores on the full Herbarium 2022 validation set (167,955 images, 15,334 species), after 50 epochs at 384 px. Seed 42 except where noted. Gaps to ArcFace are in points, with 95% paired image-bootstrap intervals in brackets.

Loss	Top-1	Macro-F1	Top-5	Δ Top-1	Δ Macro-F1
DBM	**0.8443**	**0.8161**	**0.9617**	+1.45 [1.30, 1.58]	+1.89 [1.70, 2.14]
KappaFace	0.8415	0.8150	0.9569	+1.16 [1.02, 1.29]	+1.77 [1.55, 1.95]
Sub-center ArcFace	0.8314	0.8003	0.9481	+0.16 [0.02, 0.29]	+0.31 [0.09, 0.50]
ArcFace + TARA	0.8305	0.7986	0.9470	+0.06 [−0.08, 0.20]	+0.13 [−0.10, 0.33]
ArcFace (reference)	0.8299	0.7972	0.9466	–	–
ArcFace + ECC	0.8261	0.7934	0.9434	−0.38 [−0.52, −0.24]	−0.38 [−0.63, −0.21]
HACE	0.8212	0.7908	0.9520	−0.87 [−1.02, −0.72]	−0.65 [−0.91, −0.45]
X2-Softmax	0.3782	0.3136	0.7700	–	–
Seed 43 reruns (noise check)					
DBM	0.8444	0.8159		vs. seed 42: +0.01 / −0.02	
KappaFace	0.8414	0.8145		vs. seed 42: −0.01 / −0.05	
ArcFace	0.8288	0.7976		vs. seed 42: −0.11 / +0.04	
Earlier reference points at 224 px (not directly comparable)					
CE baseline, 224 px	0.7796	0.7455	0.9328		
ArcFace, 224 px	0.7971	0.7605	0.9337		

- **Sub-center ArcFace is a small but real gain** (+0.16 accuracy, +0.31 macro-F1, both intervals above zero).
- **TARA has no detectable effect.** Both intervals include zero.
- **ECC and HACE are worse than ArcFace.** HACE is the odd case: its top-5 (0.952) is higher than ArcFace's (0.947), so it gets the right species into its top 5 more often but ranks it first less often.
- **X2-Softmax fails** at 38% accuracy. Its learning curve rises slowly the whole time and does not plateau. A rerun at the paper's scale $s = 64$ learned even more slowly (accuracy 0.112 at epoch 6.8 and 0.140 at epoch 10.2) and was stopped, so the scale is not the cause. The geometry analysis shows feature collapse (Section 5.3).

5.2 Fine-grained results: per-species F1 and the role of training-set size

Herbarium 2022 caps species at 73 training images, but there is still a rare tail: 1,060 species have 10 or fewer training images. Figure 2 groups species by training-image count and compares mean per-species F1 across losses.

- **The gains are concentrated in the rare tail.** For species with 6–10 training images, KappaFace adds +6.3 points of per-species F1 over ArcFace and DBM +4.6. For species with 51–73 images, which are more than half of all species, both add only about +1 (KappaFace +1.0, DBM +1.2). Sub-center ArcFace adds less than 1 point in every band.

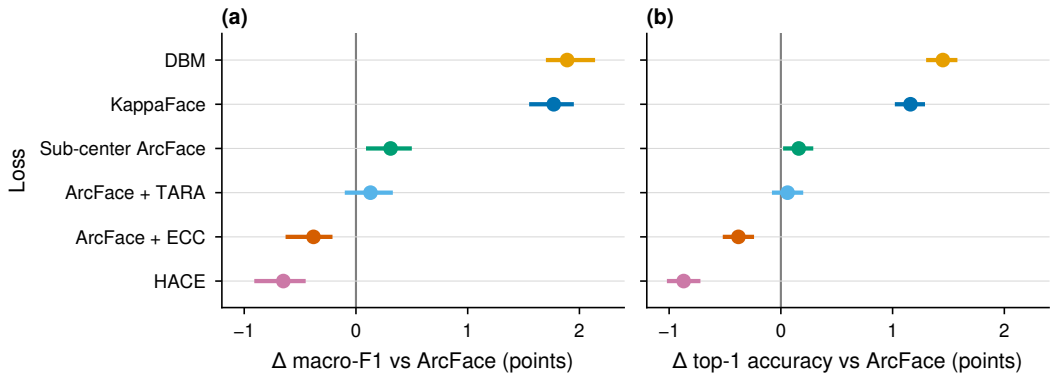


Figure 1: **Only per-class-margin losses beat ArcFace by more than a fraction of a point.** Gap of each loss to ArcFace in (a) macro-F1 and (b) top-1 accuracy, full validation set, seed 42. Bars are 95% paired bootstrap intervals over validation images, which do not include seed noise. Seed noise, measured for DBM, KappaFace and ArcFace, is about 0.1 points or less. X2-Softmax (−48 points) is off the scale.

- **Every loss still finds rare species much harder.** Even the best loss is about 24 points worse on species with 2–5 images (KappaFace 60.4) than on species with 51–73 (DBM 84.6).
- **Fewer species are never recognised.** The number of validation species with F1 = 0 falls from 367 (ArcFace) to 313 (DBM) and 255 (KappaFace).
- **No loss changes the kind of mistake.** For every working loss, 75–78% of errors are in the right genus and 87–90% in the right family (ArcFace 75.5 / 87.7%, DBM 77.8 / 89.5%). The better losses make fewer mistakes, not different ones. HACE, built around the taxonomy, does not shift this either.

KappaFace vs. DBM crossover. Figure 3 compares the two winners directly, with both seed pairs. KappaFace is better below about 10 training images: +3.4 points at 2–5 images and +2.9 at 6–10, with pooled intervals above zero. The two tie at 11–15. DBM is better from 16 images upward (−0.9, −0.4 and −0.2 points), and these bands hold most species. That is why overall macro-F1 barely separates the two.

Routing test, with no retraining. Since the two losses win in different bands, we tested a simple combination at inference time. Use KappaFace's prediction when it names a species with at most T training images, and DBM's otherwise. T was chosen on a random half of the validation images ($T = 12$ and $T = 10$ for the two seed pairs) and scored on the other half (Table 5). Routing lifts the rarest bands by 3–8 points and leaves common species within ±0.1. Only 1.4–2.2% of images take KappaFace's answer, so the overall macro-F1 gain (+0.14) is about the size of seed noise. The cost is running two models at inference time.

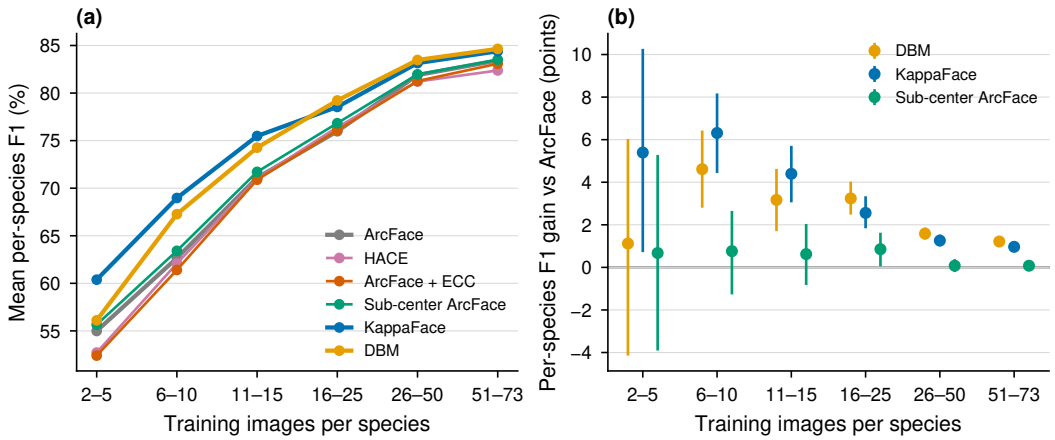


Figure 2: **DBM and KappaFace gain most on rare species; KappaFace leads on the rarest.** (a) Mean per-species F1 in bands of training images per species (104 / 956 / 937 / 1,554 / 3,710 / 8,073 species per band). (b) Paired per-species F1 gain over ArcFace; bars are 95% bootstrap intervals over the species in each band, not seed noise. Seed 42, full validation set. The 167 species with no validation images are excluded.

Table 5: **Routing rare-species predictions to KappaFace lifts the tail.** Scores on the held-out validation half, with T chosen on the other half. Accuracy and macro-F1 are fractions; band columns are mean per-species F1.

Seed pair	Model	Top-1	Macro-F1	F1, 2–5 imgs	F1, 6–10 imgs
42 ($T = 12$)	DBM	0.8449	0.8078	0.517	0.672
	KappaFace	0.8426	0.8076	0.571	0.685
	Routed	**0.8456**	**0.8092**	**0.594**	**0.700**
43 ($T = 10$)	DBM	0.8458	0.8104	0.541	0.657
	KappaFace	0.8425	0.8061	0.566	0.686
	Routed	**0.8463**	**0.8118**	**0.591**	**0.708**

5.3 Representation geometry

For every model we saved the 1024-d pooled feature of each validation image. We computed the same geometry measures as the earlier CE-vs-ArcFace study:

- *within-species cosine*: mean similarity of an image to its own species centroid;
- *nearest-species cosine*: similarity of a species centroid to the closest other centroid;
- *separation gap*: the first minus the second;
- *NC1*: within-class scatter relative to between-class scatter;
- *alignment*: cosine between each species' classifier weight vector and its feature centroid.

Higher gap and alignment and lower NC1 are "tidier" geometry in the usual sense.

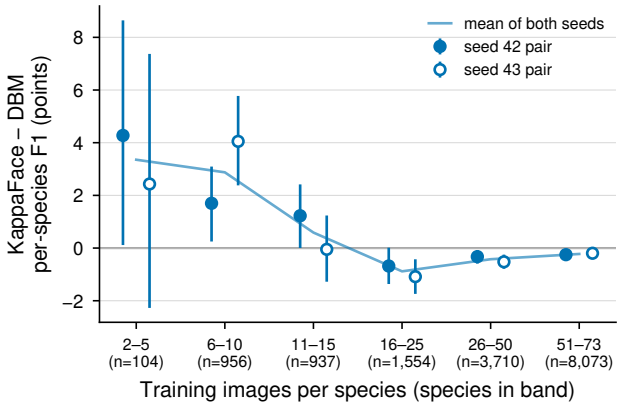


Figure 3: **KappaFace beats DBM on rare species, DBM wins on common ones.** Per-species F1 of KappaFace minus DBM, by training-image band, for two independent seed pairs (filled: seed 42, open: seed 43). Bars are 95% bootstrap intervals over species; the line is the mean of both seeds.

Table 6: **Geometry of the 384 px models** (full validation set, seed 42). The last column is the within-model Spearman correlation between per-species alignment and per-species recall.

Loss	Within	Nearest	Gap	NC1	Align.	ρ(align, recall)	Macro-F1
DBM	0.721	0.667	0.055	0.108	0.694	0.44	**0.8161**
KappaFace	0.733	0.622	0.111	0.089	0.763	0.45	0.8150
Sub-center ArcFace	0.773	0.532	0.241	0.063	0.795	0.38	0.8003
ArcFace + TARA	0.779	0.543	0.236	0.061	0.839	0.43	0.7986
ArcFace	0.772	0.530	0.242	0.062	0.838	0.44	0.7972
ArcFace + ECC	0.786	0.524	**0.261**	**0.056**	**0.840**	0.39	0.7934
HACE	0.684	0.661	0.023	0.142	0.474	0.46	0.7908
X2-Softmax	0.992	0.993	−0.001	0.017	0.610	−0.10	0.3136

Results are in Table 6 and Figure 4. The seed-43 reruns of DBM, KappaFace and ArcFace reproduce every value to within 0.003.

- **Across losses, tidier average geometry goes with *lower* macro-F1.** DBM and KappaFace have separation gaps of 0.055 and 0.111, against 0.242 for ArcFace, and lower alignment (0.69, 0.76 vs. 0.84). ECC has the tidiest geometry of any loss and the lowest macro-F1 of the ArcFace family. Together with the earlier CE→ArcFace result, where geometry improved a lot and accuracy only a little, this means average geometry is not a reliable target for choosing a loss on this data.

- **Within any one model, geometry still predicts which species are recognised.** Species whose classifier direction lines up better with their cluster have higher recall in every model (Spearman 0.38–0.46).

- **X2-Softmax features have collapsed.** Every image points in almost the same direction (cosine 0.99 to its own species and 0.99 to the nearest other species), so 15,501 species must be separated inside a tiny cone. This is consistent with its slow learning.

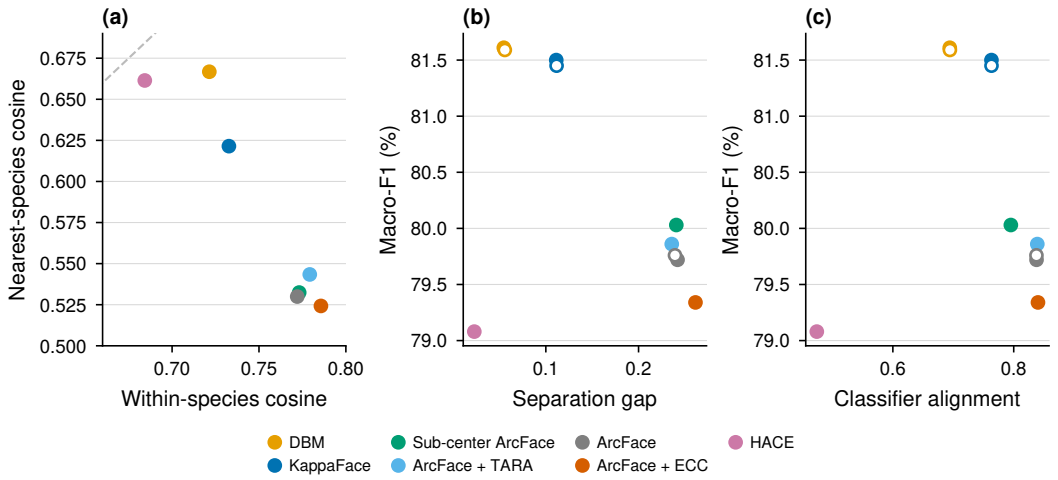


Figure 4: **The two best losses have the loosest geometry.** (a) Within-species against nearest-species cosine; points further below the dashed diagonal have species clusters that are more separated. DBM and KappaFace sit closest to the diagonal of any working loss. (b, c) Macro-F1 against separation gap and classifier–centroid alignment. Filled markers are seed 42; open markers are the seed-43 reruns, which land on top of them. X2-Softmax (macro-F1 31%, collapsed features) is off the scale.

Table 7: **DBM beats ArcFace on every rung, and its macro-F1 lead grows with the number of species.** Final scores after 131,250 steps, full validation set of each rung, one seed per run.

	DBM		ArcFace		DBM lead (points)	
Rung (epochs)	Top-1	Macro-F1	Top-1	Macro-F1	Top-1	Macro-F1
15.5k (24.6)	0.8959	0.8436	0.8842	0.8178	+1.17	+2.58
32k (20.1)	0.8882	0.7754	0.8777	0.7452	+1.05	+3.02
64k (14.9)	0.8619	0.6818	0.8490	0.6429	+1.29	+3.89

- **HACE has low species alignment (0.47).** Its classifier also has genus and family rows and its prediction is aggregated over the tree, so the species rows alone describe only part of the model.

5.4 Scaling ladder

Table 7 and Figure 5 give the ladder results.

- **The ranking holds at every size**: DBM beats ArcFace on both metrics on every rung.
- **DBM's macro-F1 lead grows with the number of species**, +2.6 → +3.0 → +3.9 points, while the accuracy lead stays flat at +1.0 to +1.3. The larger rungs add mostly rare species (Table 3). Rare species count for little in accuracy, because they have few images, but count fully in macro-F1. This matches the 15k per-species result, where DBM's gains were largest for rare species.

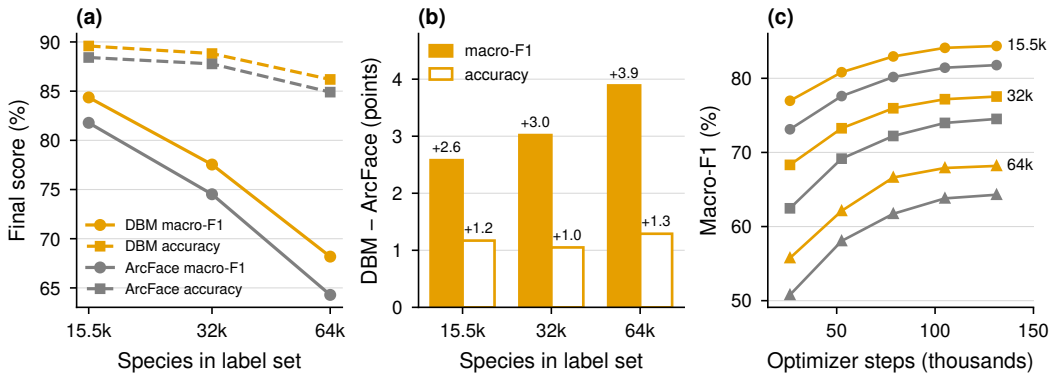


Figure 5: **DBM's macro-F1 advantage grows as rare species are added.** (a) Final macro-F1 (solid) and top-1 accuracy (dashed) of DBM (orange) and ArcFace (grey) on nested sets of 15.5k, 32k and 64k species. (b) DBM's lead over ArcFace. (c) Macro-F1 during training at the five evaluations (every 26,250 steps); colors as in (a). All runs use the same 131,250-step budget. Single seed per run; seed noise at 15k was about 0.1 points.

- **Both losses lose much more macro-F1 than accuracy as species are added**: DBM goes from 0.844 to 0.682 in macro-F1 but only from 0.896 to 0.862 in accuracy.
- **The lead appears early and narrows slowly.** At the first evaluation (20% of training) DBM leads by +3.9, +5.9 and +5.0 points of macro-F1 on the three rungs. The lead shrinks during training but never closes (Figure 5c). Both losses are still improving slowly at the end, especially on the 64k rung, which sees fewest passes over its data.

Caveats. One seed per ladder run. Seed noise at 15k was about 0.1 points, far below these gaps, but it was measured on different data. Every rung gets the same compute, so the larger rungs are less trained per image, and more training might close part of the gap between rungs. On the 64k rung the median species has only 3 validation images, so its per-species F1 values are individually noisy.

6 Summary of findings so far

1. Per-class adaptive margins (DBM, KappaFace) are the only interventions that improve on ArcFace by more than a fraction of a point: about +1.9 macro-F1 and +1.2 to +1.5 top-1, reproducible across seeds.
2. The gains sit in the rare tail. KappaFace is better below about 10 training images, DBM above about 15. A no-retraining router lifts the rarest bands by 3–8 points.
3. Hierarchy (HACE), class-center regularisation (ECC) and BioCLIP alignment (TARA) did not help as implemented. The share of within-genus errors (75–78%) is the same for every working loss.

4. Average geometry is not a reliable proxy for accuracy across losses. The best losses have the loosest species clusters.

5. On the ladder, DBM beats ArcFace at 15.5k, 32k and 64k species, and its macro-F1 lead grows with the number of species.

Not yet measured. KappaFace on the ladder, and the routing test on the ladder (runs in progress). Per-species F1 by training-image count on the ladder rungs, which needs per-image predictions to be saved for the ladder models. Geometry split by rarity band. FlexiGrad and multi-task family/genus/species heads. More than one seed for the losses other than DBM, KappaFace and ArcFace.

A Provenance

Table 8: Run identifiers behind each result (first 8 characters).

Result	Runs
Full-set re-scores of the 8 losses	`f701e70f`
Training runs (seed 42)	ArcFace `6a899d4a`, sub-center `47362b64`, HACE `677aa821`, DBM `c2fecdb6`, X2 `0b103f18`, KappaFace `f13e6236`, ECC `8fff330a`, TARA `601daea1`
Seed-43 reruns	DBM `61e3b0a7` (re-score), KappaFace `cf97fa8d`, ArcFace `f4e0099f`
Per-species F1, bootstrap, error structure	`2964fe72`
KappaFace–DBM crossover and routing	`ab4f160f`
Geometry (11 models)	`bf50bd59`
X2-Softmax at $s = 64$	`877404ac`
224 px references	CE `be436c7e`, ArcFace `be730cdf` / `da0667dc`
Ladder data build	`5c9db6fa`
Ladder final runs (DBM / ArcFace)	15.5k `dc59af2e` / `e3ecf637`, 32k `ca0355e1` / `f30277ed`, 64k `29abd9c9` / `2244e684`

Figures are generated by `figs/make_figures.py` from the data files in `figs/data/`.

References

J. Deng, J. Guo, J. Yang, N. Xue, I. Kotsia, and S. Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. arXiv:1801.07698. `https://www.alphaxiv.org/abs/1801.07698`

J. Deng, J. Guo, T. Liu, M. Gong, and S. Zafeiriou. Sub-center ArcFace: Boosting face recognition by large-scale noisy web faces. In *ECCV*, 2020. `https://doi.org/10.1007/978-3-030-58621-8_43`

C. Oinar, B. M. Le, and S. S. Woo. KappaFace: Adaptive additive angular margin loss for deep face recognition. *IEEE Access*, 2023. arXiv:2201.07394. `https://www.alphaxiv.org/abs/2201.07394`

J. Xu, X. Liu, X. Zhang, Y.-W. Si, X. Li, Z. Shi, K. Wang, and X. Gong. X2-Softmax: Margin adaptive loss function for face recognition. arXiv:2312.05281. https://www.alphaxiv.org/abs/2312.05281

M. Son, I. Koo, J. Park, and C. Kim. Difficulty-aware balancing margin loss for long-tailed recognition. In *AAAI*, 2025. https://doi.org/10.1609/aaai.v39i19.34261

H. Yao, Q. Miao, P. Zhao, C. Li, X. Li, G. Feng, and R. Liu. Exploration of class center for fine-grained visual classification. *IEEE TCSVT*, 2024. https://doi.org/10.1109/tcsvt.2024.3406443

A. Chan, D. D'Ascenzo, and S. Cultrera di Montesano. When labels have structure: Improving image classification with hierarchy-aware cross-entropy. arXiv:2605.06274. https://www.alphaxiv.org/abs/2605.06274

H. He, Z. Tan, and Y. Peng. Taxonomy-aware representation alignment for hierarchical visual recognition with large multimodal models. arXiv:2603.00431. https://www.alphaxiv.org/abs/2603.00431

S. Stevens, J. Wu, M. J. Thompson, E. G. Campolongo, C. H. Song, D. E. Carlyn, W.-L. Chao, and Y. Su. BioCLIP: A vision foundation model for the tree of life. In *CVPR*, 2024. https://www.alphaxiv.org/abs/2311.18803

Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. arXiv:2103.14030. https://www.alphaxiv.org/abs/2103.14030

R. de Lutio, D. Little, B. Ambrose, and S. Belongie. The Herbarium 2021 Half-Earth challenge dataset. arXiv:2105.13808. https://www.alphaxiv.org/abs/2105.13808